

$\mu: G \rightarrow G'$ be a group homomorphism

Def: The kernel of μ is

$$\text{Ker}(\mu) := \{g \in G \mid \mu(g) = 1_{G'}\}$$

The image of μ is

$$\text{Im}(\mu) := \mu(G) \subset G'$$

Note: $\text{Ker}(\mu) \subset G$

• Let f homomorphisms f_1 & f_2

$$G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} G_3$$

This sequence is called exact if

$$\text{Im}(f_1) = \text{Ker}(f_2)$$

$\Leftrightarrow g \in G$ then $f_1(g_1) = g_2 \in G_2$

now if $f_2 \circ f_1(g_1) = 1_{G_3}$

The exactness of the sequence, is the

statement $f_2 \circ f_1(g_1) = 1_{G_3}$

For vector spaces

$$\Rightarrow f_2 \circ f_1 = 0$$

*We can generalise this to the sequence of several groups

in homo.:

$$G_{i-1} \xrightarrow{f_{i-1}} G_i \xrightarrow{f_i} G_{i+1} \xrightarrow{f_{i+1}} \dots$$

which is exact if $\text{Im}(f_{i-1}) = \text{Ker}(f_i)$

Def: A short exact sequence:

$$1 \hookrightarrow G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} G_3 \hookrightarrow 1$$

which is exact at G_1, G_2, G_3

1: trivial group w/ one element.

• There is a unique homo. $1 \rightarrow G_1$ and $G_3 \rightarrow 1$.

* $1 \rightarrow G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} G_3 \rightarrow 1$ is equivalent to

(i) Exactness at G_1 : The kernel of f_1 is the image of the inclusion $\{1\} \hookrightarrow G_1$ and hence is the trivial group.

$$\bar{c}(1 \cdot 1) = \bar{c}(1) \bar{c}(1) = 1_{G_1}$$

$$f_1(\bar{c}(1)) = f_2(1_{G_1}) = 1_{G_2}$$

• Therefore f_1 is an injection of G_1 to G_2

(ii) Exactness at G_2 $\text{Im}(f_1) = \text{Ker}(f_2)$

Is by definition.

(iii) Exactness at G_3 : $G_3 \hookrightarrow \{1\}$

$$f_2(g_2) = g_3$$

$$\mu_3(g_3) = 1_3 \quad \dots$$

$$\text{Im}(f_2) = \text{Ker}(\mu_3)$$

* Note For any homo $\mu: G \rightarrow G'$

$$1 \rightarrow \text{Ker}(\mu) \rightarrow G \xrightarrow{\mu} \text{Im}(\mu) \rightarrow 1$$

Note $1 \rightarrow \text{Ker}(\mu)$ is trivial

$$1_G \rightarrow 1_G \text{ and } \text{Im}(\mu) \rightarrow G$$

is $1_G \hookrightarrow G$

Thm: $\text{Ker} \subset G$ is a normal subgroup of G :

$$\mu(g \cdot k \cdot g^{-1}) = \mu(g) \mu(k) \mu(g^{-1}) = \mu(g) \mu(g^{-1}) = 1_{G'}$$

* G/H and $\mu(G)$ are both groups.

these two groups are closely related

$$\mu(g) = \mu(g') \Leftrightarrow gk = g'k$$

$$\text{Thm. } \mu(G) = G/\ker(\mu)$$

Proof: We associate the coset gK to $\mu(g)$ in G'

$$\psi : gK \mapsto \mu(g)$$

(i) ψ is well defined:

$$gK = g'K \Rightarrow \exists K \in K$$

$$g'K = gK$$

$$\mu(g'K) = \mu(gK)$$

$$\mu(g')\mu(K) = \mu(g)\mu(K) \Leftrightarrow \mu(g) = \mu(g')$$

(ii) ψ is a homomorphism:

$$\psi(g_1K, g_2K) = \mu(g_1K \cdot g_2K) = \mu(g_1g_2K)$$

(iii) Surjection clear:

$$\text{Assume } \mu(g') = \mu(g)$$

$$\text{then } \exists K \text{ s.t. } g' = gK \Leftrightarrow g'K = gK.$$

From the above theorem,

$$I \hookrightarrow K \hookrightarrow G \rightarrow G/K \rightarrow I$$

$$\Rightarrow \text{any } I \rightarrow N \rightarrow G \rightarrow Q \rightarrow I$$

* N is isomorphic to a normal subgroup of G

$$Q \cong G/N$$

Q : Quotient Group

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